

Ritik Gupta

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PROFILE

AI/ML and Backend Python Developer with 3 years of experience building machine learning models and scalable backend systems. Proficient in Python, FastAPI, scikit-learn, TensorFlow, and PostgreSQL. Skilled in API development, data processing, model deployment, and cloud services like AWS. Strong problem-solving abilities and a passion for delivering end-to-end AI-driven solutions.

PROFESSIONAL EXPERIENCE

AI/ML Engineer, Mobcoder

09/2024 – present | Noida

- Designed and delivered **end-to-end AI systems** for real-world business use cases, driving projects from **POC to production**, including system design, model selection, backend integration, and deployment, improving solution delivery efficiency and scalability.
- Developed **production-grade backend services using Python and FastAPI**, implementing **user management, API contracts, and business logic**, while orchestrating **LLM and ML workflows** to enable scalable AI-driven applications.
- Architected and prototyped **Generative AI solutions**, including **RAG pipelines, vector database integrations (e.g., Qdrant/Vertex AI), and agentic workflows**, ensuring high reliability, modular design, and production readiness.
- Implemented **secure and scalable backend architectures**, incorporating **JWT authentication, RBAC, secure hashing, input validation, CORS policies, and rate limiting**, enhancing system security and reducing vulnerability risks.
- Evaluated multiple architectural approaches and contributed to **data-driven technical decisions**, optimizing for **cost, scalability, performance, and delivery timelines**, resulting in more efficient and maintainable systems.

Software Engineer, Ionicx Digital

08/2023 – 08/2024 | Chennai

- Developed and maintained a **RESTful API platform** for excursion booking and travel management systems, enabling seamless end-to-end user interactions and improving operational efficiency.
- Engineered scalable and robust **API endpoints for user authentication, trip creation, scheduling, and real-time data retrieval**, ensuring smooth integration with frontend applications and enhancing system responsiveness.
- Implemented **secure data handling and validation mechanisms**, following industry best practices to strengthen application security, reduce vulnerabilities, and ensure high data integrity.
- Designed the system using **object-oriented programming (OOP)** and **modular architecture principles**, improving code maintainability, scalability, and supporting rapid feature enhancements.

PROJECTS

Akari-bot

- Built a **FastAPI-based AI backend** for health and knowledge workflows, integrating **JWT authentication, Redis caching, MongoDB persistence, and structured API routing**.
- Implemented **Retrieval-Augmented Generation (RAG)** using **Qdrant** with **Semantic Scholar ingestion** and **prompt-driven LLM orchestration** across **OpenAI, Gemini, Anthropic, and Mistral**.
- Designed **chart-generation services** that convert **natural-language queries into validated MongoDB aggregation pipelines**, returning **chart-ready datasets**.
- Developed **claim verification, content pipeline, and safety/crisis-classification modules** with **strict grounding rules for LLM responses**.
- Strengthened production reliability through **rate limiting, retry mechanisms, and multi-stage validation** to ensure **data integrity and system stability**.

Prediction and Recommendation Analysis

- Developed a **predictive analytics module** using **Machine Learning** algorithm **RandomForestRegressor** to accurately forecast metrics such as call volume, agent idle time, and average handling time, improving resource allocation.
- Created a **user-friendly frontend** with **React.js**, allowing users to **visualize predictions** and recommendations in real time.
- Implemented a background scheduler using Python's **threading** module for timely evaluations, allowing the system to trigger web notifications via **Google Firebase Cloud Messaging (FCM)** based on threshold conditions and recommendations, ensuring real-time communication and user engagement.
- Applied industry-standard methodologies for predictive accuracy, utilizing **Python, scikit-learn, Pandas and NumPy** ensuring reliability in performance analytics.

SKILLS

Machine Learning & AI Frameworks

Generative AI, Agentic AI, Retrieval-Augmented Generation (RAG), Prompt Engineering, AI Agents, Multi-Agent Systems, LangChain, LangGraph, LlamaIndex, MCP (Model Context Protocol), OpenAI API, Gemini API, Anthropic Claude API, Hugging Face Transformers, TensorFlow, PyTorch, Scikit-learn, Embeddings, Semantic Search, Hybrid Search, Model Evaluation, Fine-tuning Fundamentals

Backend Technologies

Python, FastAPI, Django, RESTful APIs, GraphQL, WebSockets, SQLAlchemy, Pydantic, Authentication & Authorization (JWT, RBAC), API Security, Rate Limiting, Input Validation, CORS, Microservices Architecture, Async Programming (asyncio), Background Tasks

Database

PostgreSQL, MySQL, MongoDB, SQLite, Redis, Qdrant, Pinecone, Vertex AI Vector Search

Cloud & DevOps

AWS (EC2, S3, IAM, Lambda, API Gateway), Google Cloud Platform (GCP), Docker, CI/CD Pipelines, Git, GitHub, Gitlab, Linux

CERTIFICATES

- Python Programming
- MySQL
- Enhancing and Soft Skill Personality

EDUCATION

Bachelor Of Technology(CSE), Apollo Institute of Technology

07/2019 – 06/2023